

MAT 426, Class 13: Convergence of QR Iteration

CONVERGENCE OF QR ITERATION Suppose that A is an $n \times n$ matrix that has eigenvalues $\lambda_1, \dots, \lambda_n$ with $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$. Let X denote the matrix whose i th column is an eigenvector of A corresponding to λ_i and suppose that X^{-1} has an LU decomposition. Then, the subdiagonal entries of the matrices A_k tend to zero and for $i = 1, 2, \dots, n$, $(A_k)_{ii} \rightarrow \lambda_i$.

Idea of Proof:

- Existence of X^{-1} :
- Orthogonal Similarity to A :
- Diagonalization of A :
- Sufficiency of showing that $\lim_{k \rightarrow \infty} P_k = Q$:

- Why does $\lim_{k \rightarrow \infty} P_k = Q$?

- Why does $\lim_{k \rightarrow \infty} D^k L D^{-k} = I$

Two facts QR Factorization using Gram-Schmidt Orthogonalization:

- The number of FLOPS required to find the QR Factorization using Gram-Schmidt orthogonalization is _____.
- The algorithm is _____ and can result in possible _____ loss of _____.

Definition. A matrix of the form

$$H =$$

where $\mathbf{u}$ is a _____ vector in $\mathbb{R}^n$ is called a **Householder matrix**.

Example 1. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

(a) Find the associated Householder matrix H .

(b) For any $\mathbf{x} \in \mathbb{R}^2$, compute $H\mathbf{x}$.

(c) Sketch the vectors $\mathbf{u}$, $\mathbf{x}$, and $H\mathbf{x}$ on the same set of axes.

PROPERTIES OF A HOUSEHOLDER MATRIX Let $H = I - \frac{2\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}}$ be a Householder matrix with $\mathbf{u} \in \mathbb{R}^n$. Then,

(a) H is symmetric; that is $H^T = H$

(b) H is orthogonal

(c) $\|H\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$

(d) $H^2 = I$

(e) $H\mathbf{u} = -\mathbf{u}$

(f) $H\mathbf{v} = \mathbf{v}$ if $\mathbf{v}^T\mathbf{u} = 0$

(g) If $\mathbf{u}$ is parallel to a vector $\mathbf{x} - \mathbf{y}$, where $\mathbf{y} \neq \mathbf{x}$ but $\|\mathbf{y}\| = \|\mathbf{x}\|$, then $H\mathbf{x} = \mathbf{y}$.

Proof.

When computing matrix-vector and matrix-matrix products involving a Householder matrix H , the matrix H does not have to be explicitly formed since

- $H\mathbf{x} =$
- $HA^T =$
- $AH =$

Here $\mathbf{x} \in \mathbb{R}^n$, A is an $m \times n$ matrix, and H an $n \times n$ Householder matrix. Also, $\beta = \frac{2}{\mathbf{u}^T\mathbf{u}}$.